

Homework 2

ASEN 5335 Aerospace Environment

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Problem 1: Plotting the Ionosphere

(i.)

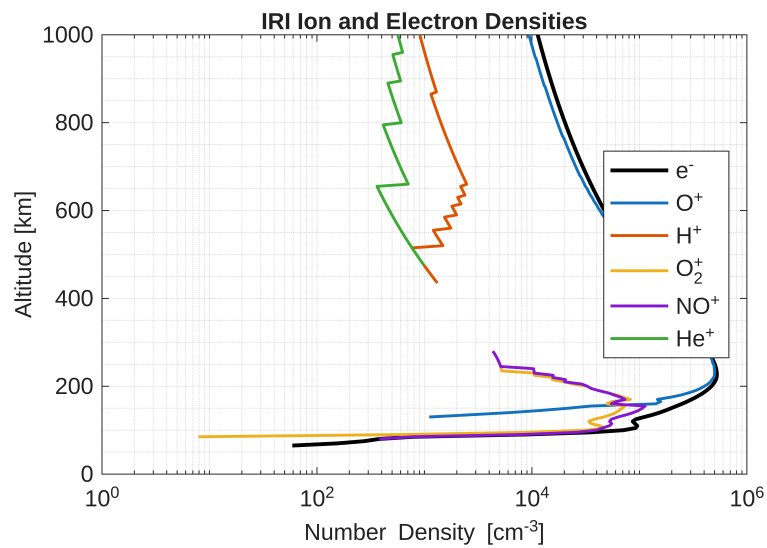


Figure 1: IRI Output

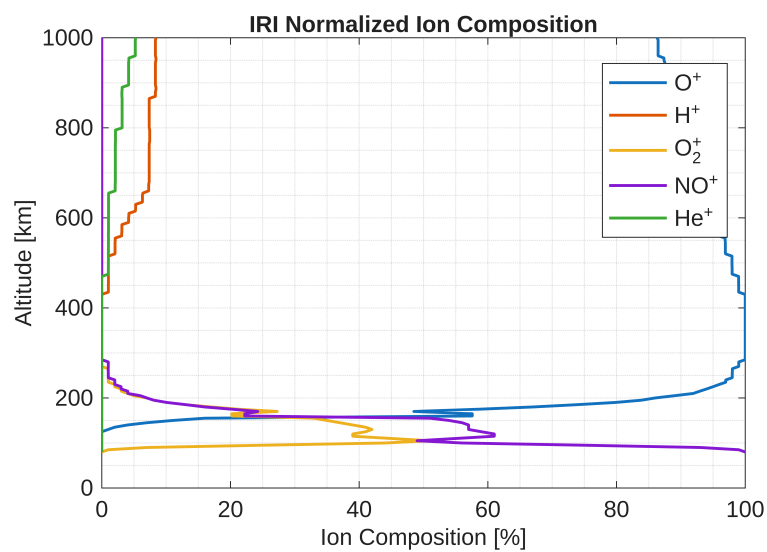


Figure 2: IRI Output Normalized

(ii.)

Below 500 km, we can observe that NO^+ dominates between 0-160 km. O_2^+ contributes more than 20% at between 90 and 175 km.

(iii.)

By convention 1 TECU is 10^{16} el/m². Integrating the total electrons from the IRI output, we find the TEC of the entire IRI output to be 12.47 TECU. When the peak electron density is 525696 m^{-3} , we can integrate the topside and bottomside separately to find they are 3.55 TECU and 8.92 TECU respectively. Normalizing against the total TEC found, we find they are 28.49% and 71.51% respectively.

Problem 2: Peak Ionization

(i.)

E Region

We are able to use the Chapman relation equation,

$$n_e(z, \theta) = \left(\frac{q_{max}}{\alpha_{eff}} \right)^{1/2} \exp \left[\frac{1}{2} (1 - z_1 - e^{-z_1}) \right]$$

Solving for q_{max} ,

$$q_{max} = \alpha_{eff} n_e^2 \exp[-(1 - z_1 - e^{-z_1})]$$

At the peak altitude $z = z_{max}$, $z_1 = 0$, so the exponential term is 1 and $n_e = n_{e,max}$:

$$q_{max} = \alpha_{eff} n_{e,max}^2$$

$$q_{max_E} = (10^{-8})(1.5 \times 10^5)^2$$

$$\boxed{q_{max_E} = 225 \text{ cm}^2/\text{s}}$$

F1 Region

Repeat the calculation above for the F1 Region,

$$q_{max_{F1}} = (7 * 10^{-9})(2.5 \times 10^5)^2$$

$$\boxed{q_{max_E} = 437.5 \text{ cm}^2/\text{s}}$$

F2 Region

Repeat the calculation above for the F2 Region,

$$q_{max_{F2}} = (5 * 10^{-9})(5.0 \times 10^5)^2$$

$$\boxed{q_{max_{F2}} = 1250 \text{ cm}^2/\text{s}}$$

(ii.)

The reason why the ionization peak altitude z_{max} moves upwards with increasing solar zenith angle χ when all other factors are kept constant is because when examining the equation,

Problem 3: Spacecraft Charging in LEO

Problem 4: Spacecraft Charging in GEO

Problem 5: Radio Wave Propagation

Problem 6: Radio Blackout

Problem 7: GPS Delays

Problem 8: D Region Signal Attenuation

Problem 9: ICON Mission